

Exercise sheet 0

PRISM - Curves and Surfaces

1. How can you define a curve mathematically? Think of different ways to do so.
2. One way to define a curve is to “trace” it with a parametrization $f(t) = (x(t), y(t))$ defined on some interval. For instance the line $x = y$ can be traced by the parametrization (t, t) . The parametrization is a way to “draw” the curve. But there may be other ways to parametrize the same curve. The same line $x = y$, for example, can be traced using $(2t, 2t)$.
 - (a) Find a parametrization for any given line given by $y = mx + b$
 - (b) Find a parametrization for the circle of radius r .
 - (c) Find a parametrization for the general ellipse, parabola, and hyperbola.
3. We have an intuitive idea of curvature. We all know what it means for a curve to be quite curved vs less curved. We want to come up with a mathematically precise definition that captures that intuitive notion. When we finally define curvature, we can test it by seeing if it gives what we expect in obvious cases like a line or a circle. What do you think the “curvature” of a line should be? What about the curvature of a circle of radius r ?
4. If a car is moving along a road, how would you figure out from the velocity of the car if the road was curved and by how much? Can you use this to come up with a concept of curvature? Write it mathematically using a parametrization of a curve. Will you use a particular kind of parametrization? Test it to see if it gives you the correct curvatures for the line and the circle of radius r .
5. Draw a curve (let us call it C) on a sheet of paper. At a particular point p of the curve, draw a small circle which is tangentially touching the curve at that point. Now draw a very big circle touching C tangentially at p . Can you see a difference between the way the curve is tangentially touching the bigger circle vs the way it is tangentially touching the smaller circle? Can you use this to identify the circle with the best possible fit? Can we not define the curvature of the curve as the curvature of the best fitting circle, which you have already defined in question 3. Do you think this definition matches with the one you came up with in question 4? Later, we will answer this question using maxima and minima!